

CoRoT-1b

R-2649

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-1_130126 · created 2026-08-29T23:16:13+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456318.7902 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.142 +/- 0.041
Transit depth (Rp/Rs)^2	2.03 +/- 1.17 [%]
Orbital Inclination (inc)	87.62 +/- 2.91
Airmass coefficient 1 (a1)	7.43 +/- 3.38
Airmass coefficient 2 (a2)	-1.36 +/- 0.64
Scatter in the residuals of the lightcurve fit is	50.3 %
Best Comparison Star	None
Optimal Aperture	1.34
Optimal Annulus	6.25
Transit Duration (day)	0.1103 +/- 0.0077

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.142 $\pm$ 0.041 vs 0.1388 (deviation 0.08 σ)
Duration fit vs published	0.1103 d vs 0.1046 d (deviation 0.74 σ)
Mid-time O–C	4.1 min
Depth SNR	3.5
Residual scatter	50.3 %

Light curve

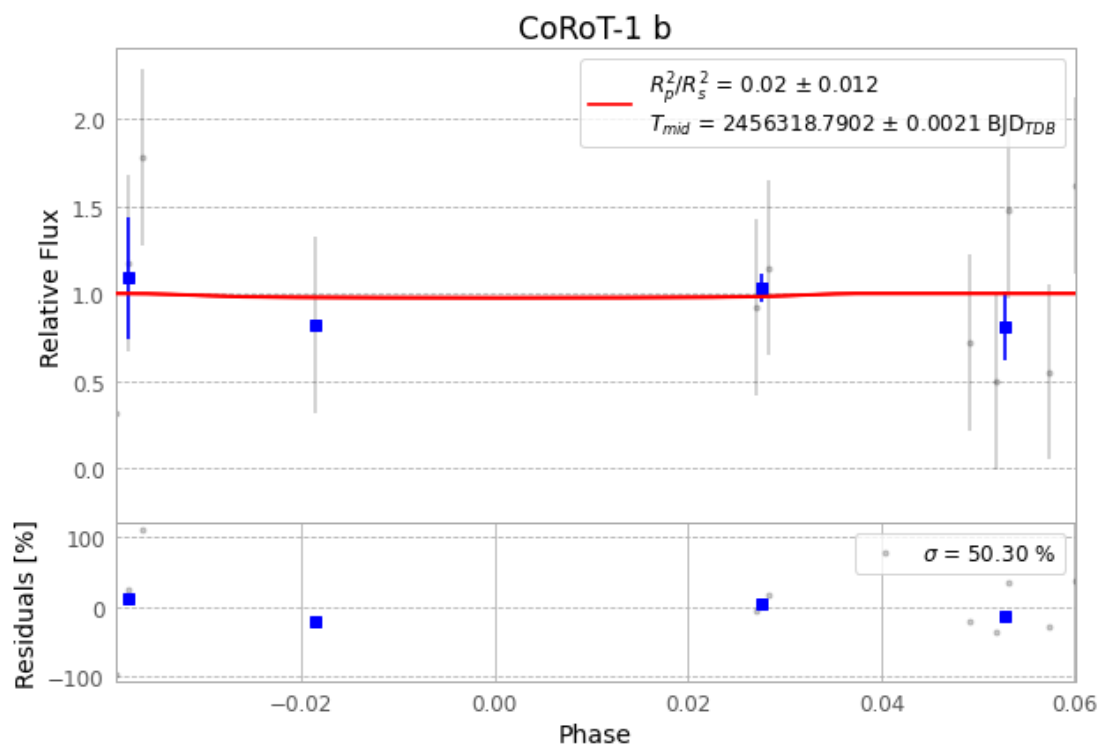


Figure 1 — FinalLightCurve_CoRoT-1 b_2013-01-25.png (SHA-256 d8159b193191c7d8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [254, 249], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d0c85b9c26dea53d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

77 FITS frames (51 MB), CoRoT-1130126051512.FITS → CoRoT-1130126090312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-1-130126.log (637b8cb8760e...), FinalLightCurve_CoRoT-1 b_2013-01-25.png (d8159b193191...), FinalLightCurve_CoRoT-1 b_2013-01-25.csv (10794dd0cacb...)

Compute resources

Wall time 133.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-29 23:16Z.